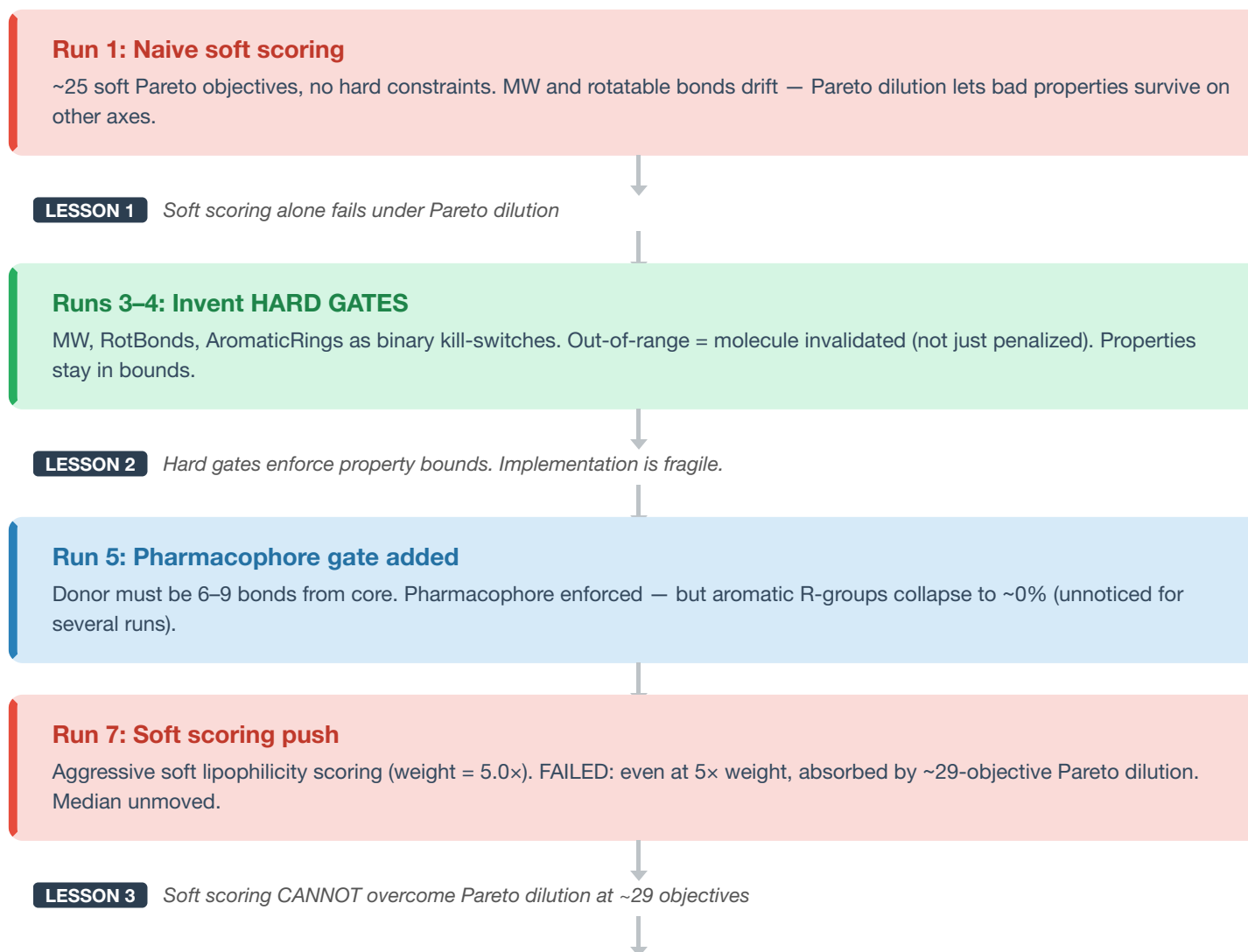


# Figure 1: How the Agent Learned RL Scoring Strategies

14 experiments over ~10 days on a real generative chemistry campaign



### Run 8: Hard gate + cold start

Hard lipophilicity gate + COLD START (untrained prior). FAILED: 1–4/128 molecules survive the 5-gate stack. Agent starved.

**LESSON 4** *Cold start on tight gate stack starves the agent*

### Run 8b: Hard gate + warm start ✓

Same hard gate + WARM START from Run 7 checkpoint. SUCCESS: +1.85 unit lipophilicity shift. 42→86/128 survivors in 5 iterations.

**LESSON 5** *Warm-start + one new gate = the working recipe*

### Run 9: Composable gates ★ — Campaign Deliverable

Add metabolic stability gate, warm from 8b. SUCCESS: 5 simultaneous hard gates held. All properties in target windows.

**LESSON 6** *Hard gates are composable with proper warm-start chains*

### Runs 10–10e: Five attempts to reduce MW

v10: ceiling 580 → COLLAPSED v10b: ceiling 600 → COLLAPSED v10c: ceiling 640 → unchanged v10d: anneal → 100% killed v10e: weight=20 ("nuclear option") → MW 625.7, IDENTICAL to Run 9

**LESSON 7** *Some properties are TOPOLOGY-LOCKED (not free parameters)*

### Run 13: Shape–MW correlation test

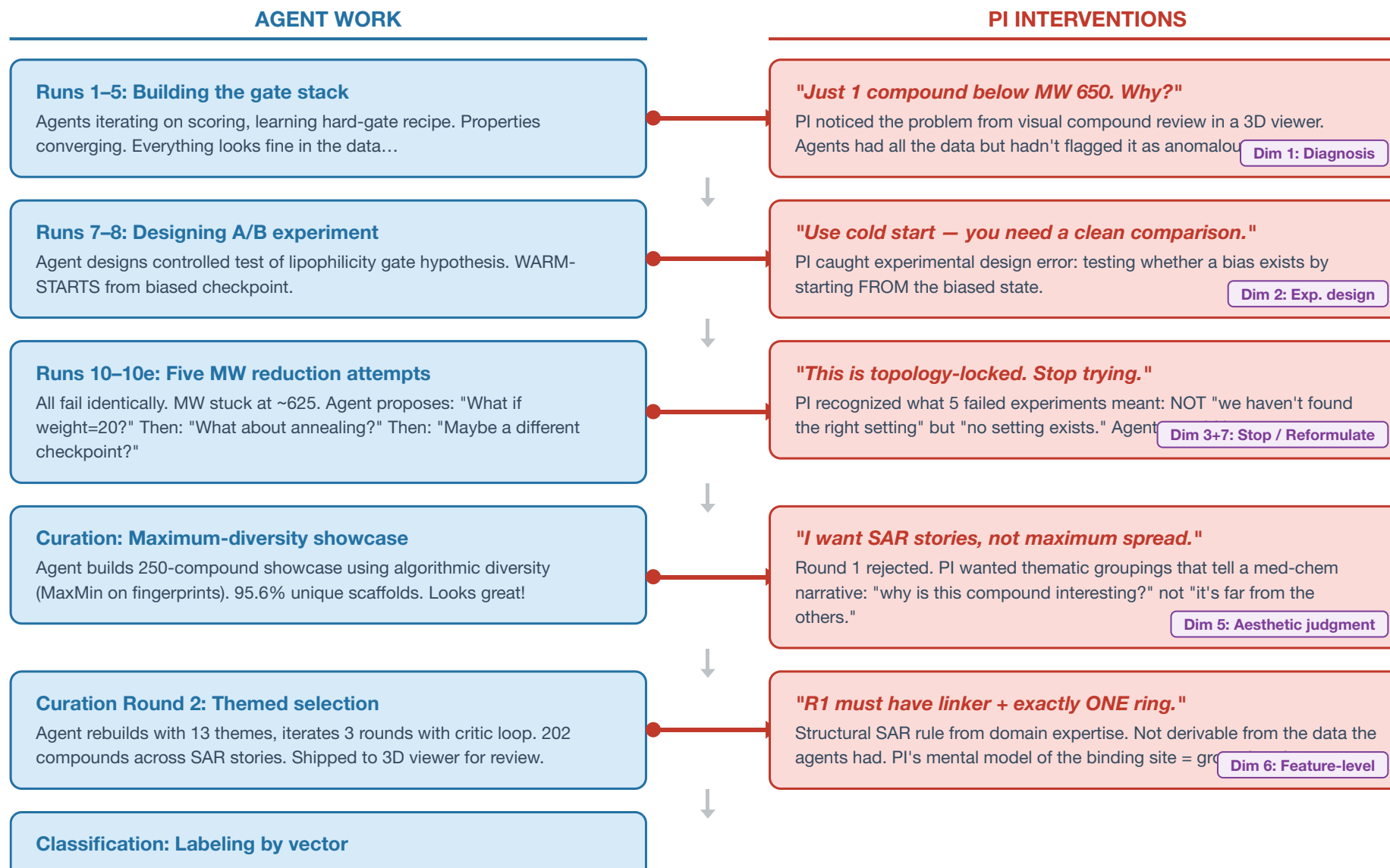
Test if 3D shape scoring drives MW up (weight 5.0). MW WORSE (687). Shape creates MW↔volume correlation — reweighting can't break it.

**LESSON 8** *Correlated objectives can't be down-weighted — remove the source*



## Figure 2: Where the PI's Input Changed the Course

Same campaign — only the moments where human expertise was essential



Agent labels compounds by which structural vector was modified, using run/file names as proxy. 48 "triple-modified" compounds found!

***"Those labels are WRONG. Decompose the structure, don't parse filenames."***

PI caught systematic classification error. 48 "triples" were actually 34 single + 10 double + 0 triple. Propagated through entire

Dim 4: Shared state

## THE BOTTOM LINE

AGENTS ran 14 RL experiments, predicted ADME on >60,000 compounds, docked thousands, curated selections, and maintained a lab notebook.

THE PI made ~6 critical interventions that each changed the trajectory. Without them: still running MW experiments on a topology-locked scaffold, with wrong labels, showcasing diverse but chemically uninteresting sets.

**Agents did 95% of the work**

**PI made 95% of the decisions that mattered**

### Benchmark dimensions:

1 — Diagnosis under Pareto complexity

2 — Experimental design with controls

3 — Knowing when to stop

4 — Multi-agent shared state

5 — Aesthetic / qualitative judgment

6 — Feature-level generalization

7 — Reformulation vs. perseverance